

YASH PALLAV PATHAK

Full Stack Developer | AI/ML Engineer | DevOps Specialist

New Delhi, India | +91 9571209434 | yash.pathak2301@gmail.com | [linkedin.com/in/yashpallavpathak](https://www.linkedin.com/in/yashpallavpathak)

PROFESSIONAL SUMMARY

Results-driven Full Stack Developer, DevOps Specialist, and AI/ML Engineer with **3+ years of professional experience** architecting, securing, and deploying enterprise-grade web, cloud, and security systems. Co-founder of **SolDevPath**, spearheading engineering delivery, technical client consultations, and proprietary product development (including SolScan and SolAmi). Proven track record executing large-scale international enterprise client engagements across the US, UK, Africa, and India—including client deployments for **Tedekstra**, **NeZaaka**, and **Bangre-Nooma**. Deep technical expertise in DevSecOps automation, GDPR/SOC compliance, container orchestration, microservices, and AI/LLM pipelines. Published IEEE researcher in AI and UAV image processing.

TECHNICAL SKILLS

AI & Machine Learning	LLM, RAG, LangChain, LangGraph, OpenAI API, Hugging Face, TensorFlow, PyTorch, Scikit-learn, Computer Vision, NLP
AI Agents & SLMs	Autonomous Agents, Multi-Agent Systems, CrewAI, Small Language Models, Model Fine-tuning, LoRA, QLoRA, Model Quantization
Model Deployment	ONNX, TensorRT, vLLM, Ollama, Docker, Kubernetes, AWS SageMaker, Hugging Face Inference API, Model Serving
Frontend	React, Next.js, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, Redux, Zustand
Backend	Node.js, Express, Java, Python, REST APIs, GraphQL, WebSockets, Microservices
Database	MongoDB, PostgreSQL, MySQL, Redis, Pinecone, ChromaDB, Vector Databases
DevOps & Cloud	AWS (EC2, S3, Lambda, RDS), Microsoft Azure (Azure VMs, Blob Storage, Azure Functions, Azure App Services, AKS, Azure DevOps, Entra ID / Azure AD), Docker, Kubernetes, CI/CD, GitHub Actions, Terraform, Linux, Nginx, Serverless
Cybersecurity	DevSecOps, GDPR Compliance, VAPT, Penetration Testing, Vulnerability Assessment, Security Auditing, OWASP Top 10, NIST, Privado, Semgrep, OWASP ZAP
Tools & Platforms	Git, GitHub, Firebase, Supabase, Postman, Figma, Streamlit, SEO Optimization

PROFESSIONAL EXPERIENCE

Co-Founder & Technical Delivery Lead | SolDevPath

New Delhi, India | 2025 – Present

- Executive Leadership & Client Delivery:** Co-founded and lead technical engineering delivery, client discovery, solution architecture, and end-to-end execution of enterprise software for international clients across the US, UK, Africa, and India.
- SolScan DevSecOps Platform Engineering:** Architected and built **SolScan**, a local-first DevSecOps and privacy-scanning CLI orchestrating Privado (GDPR), Semgrep (SAST), OSV Scanner/npm audit (SCA), and OWASP ZAP (DAST) with consolidated Markdown reporting and organization-aware API key CI gates.
- SolAmi Digital Health Platform:** Designed and engineered **SolAmi**, an AI-powered 3D virtual therapeutic tutor platform featuring real-time biometrics, gaze/focus analytics, voice cloning via ElevenLabs, and HL7 FHIR EHR data synchronization.
- International Client Deployments:** Architected, built, and launched production platforms for global clients including **Tedekstra** (UK), **NeZaaka** (West Africa), and **Bangre-Nooma**, delivering scalable cloud hosting, secure API pipelines, and high-performance frontend interfaces.

- **Applied GenAI & Cloud Infrastructure:** Built custom enterprise AI workflows including production RAG pipelines, Small Language Model (SLM) deployments, and LLM automation using LangChain and LangGraph alongside containerized microservices and CI/CD pipelines.

DevOps Consultant & DevSecOps Lead (Client Engagement) | [Tedekstra \(tedekstra.com\)](https://tedekstra.com)

Liverpool, UK | 2025 – Present

• Enterprise DevSecOps & Security Gate Automation (SolScan Integration):

- Integrated and rolled out the **SolScan** CLI pipeline across core Tedekstra repositories (including internal platforms like total-office).
- Enforced shift-left security policies by establishing organization-aware pre-scan API authorization gates (expectedOrg: "Tedekstra" via bearer token validation).
- Automated continuous static application security testing (SAST) using **Semgrep** (p/ci, p/security-audit, p/owasp-top-ten, p/secrets) wired with SEMGREP_APP_TOKEN in GitHub Actions for supply chain protection.
- Integrated automated Software Composition Analysis (SCA) leveraging **Google's OSV Scanner** (with fallback to npm audit) to catch package vulnerabilities before code reaches staging.
- Implemented baseline and authenticated dynamic application security testing (DAST) using containerized **OWASP ZAP** automation scripts (zap-automation.yaml), configuring spider depths, session states, and custom auth endpoints to test running services.

• Product Delivery, Quality Engineering & Release Governance:

- Spearheaded the end-to-end operational release and deployment lifecycle for major product rollouts, including the **MacroTrack Operations module**.
- Architected and maintained comprehensive **Playwright** end-to-end (E2E) automated regression testing suites, ensuring zero-regression deployments across multi-browser enterprise environments.
- Governed production release gates, staging sanity verification, containerized rollouts, and rollback procedures.

• Penetration Testing & Security Vendor Coordination:

- Directed trial external penetration testing and vulnerability assessments for the **Nori HR** application.
- Managed end-to-end engagement logistics with third-party cybersecurity auditors: defined scope boundaries, provisioned isolated test environments, coordinated privileged credentials, and triaged discovered findings into remediation roadmaps.

• Internal Automation & Developer Productivity:

- Engineered internal tooling and custom API automations (such as automated timesheet submission services and operational reporting scripts) to reduce friction and eliminate manual reporting workflows across the engineering org.
- Maintained CI/CD pipelines, Docker environments, and cloud infrastructure monitoring across active development branches.

• Community Outreach & Event Leadership:

- Represented Tedekstra externally by organizing and hosting the **HACK2UK** collegiate hackathon at GMIT College, formulating technical problem statements, managing mentorship tracks, and leading project evaluations.

Lead Full-Stack & Cloud Solutions Architect (Client Engagement) | [NeZaaka \(nezaaka.com\)](https://nezaaka.com)

West Africa / Abidjan Client Engagement | 2025 – Present

• End-to-End Multi-Tenant Marketplace Architecture:

- Architected, built, and launched the core digital rental platform for **NeZaaka**, a verified property and rental marketplace headquartered in Abidjan, Côte d'Ivoire.
- Engineered a multi-stakeholder ecosystem connecting property owners/hosts, localized field verification agents, and tenants with role-based access control (RBAC).
- Designed an end-to-end verification and listing pipeline enabling property managers to submit title deeds, IDs, and geolocated photos for physical audit before public listing.

- **FinTech & Mobile Money Escrow Rails Integration:**

- Architected and integrated localized West African payment processing rails supporting **Wave, Orange Money, and MTN MoMo**.
- Designed an automated escrow settlement workflow that securely locks rental deposits and transaction fees, automatically dispersing payouts to landlords only after tenant verification and physical key handover.
- Engineered automated SMS and webhook payment reconciliation to prevent double-spending and ensure transaction consistency during network drops.

- **Performance Engineering for Low-Bandwidth Networks:**

- Optimized platform delivery for 3G/4G mobile devices prevalent across West African mobile networks, slashing core web vital payload sizes by 45%.
- Implemented automated cloud-based image/media compression and CDN edge caching (WebP conversion, responsive image breakpoints, and lazy loading).
- Designed an offline-first client cache and lightweight search engine supporting localized multi-parameter filtering (by commune: Cocody, Marcory, Yopougon, budget, furnished vs. unfurnished).

- **Cloud Infrastructure, Security & DevSecOps Deployment:**

- Provisioned containerized production infrastructure with Docker and automated deployment pipelines via GitHub Actions.
- Implemented zero-downtime rolling updates, automated PostgreSQL database backups, and SSL/TLS cipher hardening.
- Conducted automated static security checks and dependency audits to safeguard financial transactional records, user PII, and uploaded identity documents.

- **Product Management, Pre-Sales & Localization Delivery:**

- Led end-to-end technical discovery and product roadmap sessions with founding stakeholders, translating regional real estate friction into software requirements.
- Implemented full French-language localization (i18n), regional currency handling (XOF / FCFA), and WhatsApp Business API integration for rapid tenant-landlord communication.

Founding Developer | Genius Labs

New Delhi, India | 2024 - 2025

- Built entire technology stack from scratch, architecting scalable microservices using Node.js, TypeScript, and Python
- Implemented AI-powered features using LLMs and RAG systems for intelligent document processing and search
- Designed and deployed cloud infrastructure on AWS with CI/CD pipelines, Docker containers, and automated monitoring
- Developed real-time data pipelines and event-driven architectures using message queues and stream processing
- Led technical decisions on database design, API architecture, and system scalability

Software Development Engineer I | STEM Quest Education Pvt. Ltd

Noida, India | Jul 2024 - Apr 2025

- Built and maintained web applications using Next.js and MongoDB, reducing page load times by 30%
- Designed and implemented a CRM system for workshop management using Next.js, Node.js, Express, MongoDB, and Docker
- Deployed applications on AWS EC2 with CI/CD pipelines via GitHub Actions, streamlining deployment processes
- Designed and implemented Node.js + TypeScript microservices, optimizing backend performance by 40%
- Automated workflows via GitHub Actions, dockerized services, and managed AWS EC2 deployments, improving deployment efficiency by 30%
- Created RESTful APIs to manage customer data, inventory, and workflows, reducing API response times by 20%
- Integrated Firebase authentication with Next.js for secure JWT-based user verification

- Contributed to an educational gaming platform (Scratch by MIT), enhancing frontend user engagement by 25%
- Integrated payment gateway (Easebuzz) and SMS service (Trust Signal) for secure transactions and OTP handling

Summer Trainee | Defence Research and Development Organisation (DRDO)

New Delhi, India | May 2023 - Jul 2023

- Designed and implemented a Java-based screen recording system capturing real-time video to LAN shared directory
- Built web interface for secure playback of recorded videos, enabling streaming access across LAN devices
- System used by 20+ personnel daily for training and documentation purposes

ACHIEVEMENTS & RECOGNITION

- **Hackathon Lead & Host — HACK2UK:** Organized, led, and hosted the HACK2UK collegiate hackathon at GMIT College on behalf of Tedekstra, managing challenge statements, mentoring 150+ student builders, and driving technical jury evaluations.
- **2nd Place - Agentic Research Tool Competition:** Developed an innovative agentic AI research tool, securing 2nd place in a competitive hackathon environment
- **Open Source Contributor - 2+ Years:** Active contributor to open-source projects with extensive bug hunting, PR reviews, and feature development across multiple repositories
- **Hackathon Participant:** Regular participant in technical hackathons, demonstrating rapid prototyping and problem-solving abilities

PROJECTS

CryptoAI Bridge *[AI, Web3, Blockchain, Smart Contracts]*

- Developed an innovative platform bridging AI and cryptocurrency ecosystems, enabling intelligent trading strategies
- Implemented AI-powered market analysis using sentiment analysis and price prediction models
- Built smart contract integration for automated trading execution on Ethereum and Polygon networks
- Created real-time dashboard for portfolio tracking and AI-driven investment recommendations

Precision Agriculture: AI & UAV Technology *[Python, OpenCV, TensorFlow, UAV Systems]*

- Published research paper at IEEE ICTACS 2024: "Analyzing Crop Health Using AI and UAV Technology"
- Developed AI-powered image analysis system for real-time crop health monitoring from UAV imagery
- Implemented detection algorithms for nutrient deficiencies, pest infestations, and disease outbreaks
- Created web application for automated chlorophyll and xanthophyll detection with video processing capabilities
- DOI: 10.1109/ICTACS62700.2024.10841199 | <https://ieeexplore.ieee.org/document/10841199>

Intelligent RAG Document Assistant *[Python, LangChain, OpenAI, Pinecone, FastAPI]*

- Built enterprise-grade RAG system enabling intelligent document querying with context-aware responses
- Implemented vector embeddings and semantic search using Pinecone for sub-second retrieval
- Developed conversational memory and multi-document context handling for complex queries
- Deployed on AWS with auto-scaling and load balancing for high availability

Autonomous Drone Navigation System *[Python, ROS, YOLO, Computer Vision, SLAM]*

- Developed obstacle detection and avoidance system for UAVs using YOLO-based object detection
- Implemented real-time path planning algorithms for autonomous navigation in dynamic environments
- Created simulation environment for testing navigation algorithms before hardware deployment
- Integrated SLAM for simultaneous localization and mapping in GPS-denied environments

Multi-Agent AI Workflow Orchestrator *[Python, LangGraph, CrewAI, Docker, Redis]*

- Designed multi-agent system for autonomous task delegation and execution across specialized AI agents
- Implemented agent communication protocols and shared memory for collaborative problem-solving
- Built workflow orchestration engine with state management using LangGraph and Redis
- Created monitoring dashboard for real-time agent activity tracking and performance metrics

EDUCATION

Bachelor of Technology (Computer Science & Engineering)

Guru Gobind Singh Indraprastha University | 2020 - 2024

- CGPA: 9.07/10.0 | Rank: 29th

CERTIFICATIONS & ADDITIONAL INFORMATION

- Speaker at GDG DevFest Jalandhar - Delivered technical talk on modern web development practices
- Organizer at Hack the Mountain - Coordinated technical events and mentored participants
- President, Leaf Club/Society - Led technical community initiatives and workshops